Course Logistics

Deep Generative Models

Sajjad Amini

Department of Electrical Engineering
Sharif University of Technology

My Introduction

Personal Info

- Assistant Professor at Electronics Research Institute.
- My Office: Second floor, Electronics Research Institute
- Email: s_amini@sharif.edu

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Research Field

- Trustworthy machine learning
- Generative modeling
- Speech and audio processing

Teaching Assistant

Figure: Ali Nourian (alinourian10@gmail.com)

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(alinourian10@gmail.com)

Figure: Zahra Sodagar
(zahra.sodagar@yahoo.com)

Course Timeline

numbe	date	topics	coursework	deadline
1	1403/07/01	Logistics		
2	1403/07/03	Introduction and Foundations		
3	1403/07/08	Introduction and Foundations	**HW 1 Released**	
4	1403/07/10	Deep Autoregressive Models		
5	1403/07/15	Deep Autoregressive Models		
6	1403/07/17	Deep Autoregressive Models		
7	1403/07/22	Deep Variational Autoencoders		
8	1403/07/24	Deep Variational Autoencoders		
	Thu 1403/07/26		**HW 2 Released**	**HW 1 Deadline**
9	1403/07/29	Deep Variational Autoencoders		
10	1403/08/01	Deep Variational Autoencoders		
11	1403/08/06	Normalizing Flow Models		
12	1403/08/08	Normalizing Flow Models		
	Thu 1403/08/10		**HW 3 Released**	**HW 2 Deadline**
13	1403/08/13	Normalizing Flow Models		
14	**1403/08/15**	**Midterm Exam 1 (Autoregressive, VAE)**		
15	1403/08/20	Normalizing Flow Models		
16	1403/08/22	Generative Adversarial Nets		
17	1403/08/27	Generative Adversarial Nets		
18	1403/08/29	Generative Adversarial Nets		
19	1403/09/04	Generative Adversarial Nets		
20	1403/09/06	Energy-Based Models		
	Thu 1403/09/08			**HW 3 Deadline**
21	1403/09/11	Energy-Based Models		
22	**1403/09/13**	**Midterm Exam 2 (Flow Models, GAN)**		
23	1403/09/18	Energy-Based Models	**HW 4 Released**	
24	1403/09/20	Energy-Based Models		
25	1403/09/25	Diffusion Models		
26	1403/09/27	Diffusion Models		
27	1403/09/02	Diffusion Models		
28	1403/09/04	Diffusion Models		
29	1403/10/09	Diffusion Models		
30	1403/10/11	Evaluation Methods		
	Thu 1403/10/13			**HW 4 Deadline**
	1403/11/03	**Final Exam 15:00**		

Syllabus Overview

Objectives

- Foundational understanding of deep generative models

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- Working with different types of generative models suitable for different types of modalities

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Prerequisites

First: We will have ~ 2 sections to quickly review some prerequisites in Probability and Statistics. The prerequisites are:

- Probability and Statistics
- Linear Algebra
- Machine Learning and Deep Learning

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- Presentations are tailored toward the theory

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- Homeworks: Theory Part + [Programming Part (Toy Datasets)] + [Programming Part]

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This course is NOT

- Introduction to deep learning
- Introduction to Probability and Statistics
- General purpose graphical generative models

Grade Distribution with Project (Approximately)

Activity	Percentage
Project	20
Homework 1 [DAR]	10
Homework 2 [VAE]	10
Homework 3 [FM & GAN]	10
Homework 4 [EBM & Diffusion]	10
Midterm 1	10
Midterm 2	10
Final	20

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Activity	Percentage
Homework 1 [DAR]	10
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Midterm 1	15
Midterm 2	15
Final	30

Course Policies

Extra Credit

Extra credit will be assigned to active class participation (up to 5%).

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Course Attendance

Attendance is essential for learning objectives in this course due to the extensive material we'll cover. However, there's no formal attendance mandate.

Course Policies

Late Submissions

- You can use a total of 10 permissible late days for all homework assignments.
- You can use a maximum extension of 3 days for any single homework.
- Assignment grade reduces by 0.95 compounding factor for each day overdue beyond allowed late days.
- Solutions released 3 days post-deadline; homework not accepted afterward.

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Academic Honesty Statement

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- Please avoid academic dishonesty including:
  - Cheating
  - Fabrication
  - Plagiarism
  - Facilitating Dishonesty

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- Make sure to contact me whenever you have questions regarding *Academic Honesty*

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- Just search or talk with your classmates about high-level ideas.
- All homework series, midterms and final will be precisely checked for possible cheating.

Course Materials

Books

- Murphy, K. P. (2023). *Probabilistic machine learning: Advanced topics.* MIT press.
- Bishop, C. and Bishop, H. (2023). *Deep Learning: Foundations and Concepts.* Springer.
- Tomczak, J. M. (2022). *Deep Generative Modeling.* Springer.

Online Courses

- Ermon, S. (2023). *Deep Generative Models [CS236].* Stanford University.
- Abbeel, P. and Chen, P. and Ho, J. and Srinivas, A. (2020). *Deep Unsupervised Learning [CS294-158-SP20].* University of California, Berkeley.

LMS and Contacting Me

Learning Management Systems

For this course, we will use:

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